

— DEVICE SETUP GUIDE

DRONETAG SCOUT

Everything you need to take a Scout from the box to live drone detection — written so anyone on the team can do it, even if they have never seen one before.

ETHERNET (NON-LTE)

CLOUD MODE

POE POWERED

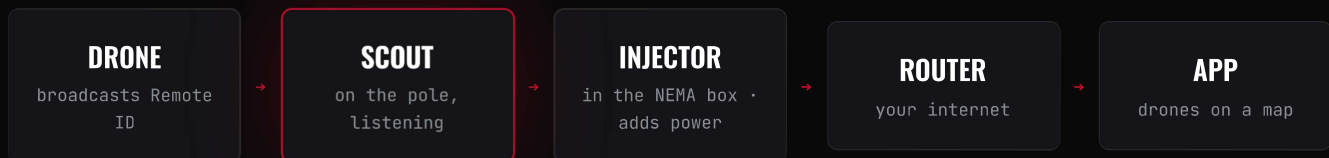


HOW THE SCOUT WORKS

Most drones constantly broadcast a radio signal called **Remote ID** — think of it as a name tag shouted into the air. It carries the drone's identity, position, height, speed, and where the pilot is standing.

The Scout is a listening post. It hears those name tags and sends them over the internet to the Dronetag cloud, and you watch every drone appear live on a map in the Dronetag App. One Ethernet cable carries both its power and its data — that trick is called **PoE** (Power over Ethernet).

SIGNAL PATH — DRONE TO YOUR PHONE



One outdoor Ethernet cable connects the Scout to the injector — it carries power up and data down. Unless the customer provides a dedicated PoE cable from indoors, the injector, router, and networking gear live in a weatherproof NEMA enclosure with power at the site.

WORDS YOU'LL SEE IN THIS GUIDE

Remote ID — the radio "name tag" drones broadcast. No Remote ID, no detection.

IP address — the Scout's house number on your network, like 192.168.1.42.

RJ-45 — the plastic clip-in plug on the end of an Ethernet cable.

PoE injector — the small box that pushes power into the Ethernet cable. Lives in the NEMA enclosure — or indoors, if the customer provides the PoE run.

Management page — the Scout's own settings website. You open it in a normal browser.

Cable gland — the screw-tight fitting where the cable enters the Scout. Keeps water out.

BEFORE YOU START

WHAT YOU NEED

Open the box and check every part against this list before you go anywhere near a ladder. If anything is missing, stop and call for help (back page).

IN THE SCOUT BOX

- Scout receiver (white weatherproof unit)
- Antennas (screw onto the top)
- Pole clamp kit — 2 clamp pieces, M6×110 and M6×30 bolts, washers, nuts
- PoE injector (power box, 120 V US outlets OK)
- Cable gland + PVC tape

TOOLS YOU BRING

- Two 10 mm wrenches
- RJ-45 crimp tool + spare RJ-45 plugs
- Outdoor-rated Ethernet cable, CAT5 or better, long enough to reach the NEMA enclosure
- Ladder + fall protection for roof work

NETWORK

- Router with a free Ethernet port and internet (lives in the NEMA enclosure)
- Laptop on the same network (for setup)

SITE

- A pole or mast, 45–110 mm across (about 1.8–4.3 inches)
- A high spot with open sky in every direction
- Power at the NEMA enclosure location — or a customer-provided PoE cable from indoors



— READ FIRST

SAFETY FIRST



NEVER WORK THE ROOF ALONE

Use fall protection and keep a spotter on the ground.



UNPLUG BEFORE YOU WIRE

The injector stays out of the wall until every cable is connected.



KEEP THE INJECTOR DRY

It is NOT waterproof. It lives inside the sealed NEMA enclosure (or indoors, if the customer provides the PoE run) — never exposed.



WATCH FOR POWER LINES

Look up before raising any pole, mast, or ladder.



FOLLOW LOCAL ELECTRICAL RULES

Dronetag requires local regulations for electrical, surge, and lightning protection to be followed.



ONLY USE THE DRONETAG INJECTOR

A different injector with the wrong voltage can damage the Scout.

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STEP 1

TEST IT INDOORS FIRST

Only if you have a network to test on — a local network or a Starlink Mini you can plug into before you go. No network available beforehand? Skip this step and run the same test on site, once the Scout is cabled to the network there.

- 01 Screw the antennas onto the Scout, hand-tight. Match each antenna to its labeled port.
- 02 Run an Ethernet cable from the Scout to the injector's **PoE / OUT** port. This is the cable that will carry power.
- 03 Run a second Ethernet cable from the injector's **LAN / DATA** port to a free port on your router.
- 04 Plug the injector into a wall outlet. The Scout has no power button — it simply starts. Give it 1–2 minutes to boot.
- 05 Leave it running — the next two steps (finding it and logging in) happen while it sits on the desk.

WHY IT MATTERS

Dronetag says to verify the device on your local network *before* physical deployment. Fixing a problem at a desk beats fixing it on a roof.

BENCH TEST — HOW IT PLUGS TOGETHER

SCOUT

on the desk

PoE
→

INJECTOR

OUT / LAN

data
→

ROUTER

internet

Wi-Fi
→

LAPTOP

same network

STEP 2

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FIND THE SCOUT'S ADDRESS

Every device on a network gets an **IP address** — a house number, like 192.168.1.42. When the Scout powers on, it asks your router for one automatically. You need that number to open its settings page.

- 01 **Easiest way:** open your router's admin page and look at the list of connected devices. Find the one named "Dronetag Scout" and write down its IP address.
- 02 **No router access?** Use a free network discovery app on your laptop or phone (e.g. Bonjour Browser on Mac — look under _https._tcp). The Scout shows up by name with its IP next to it.
- 03 Take a picture of the address or save it in the notes app on your phone. You will use it again.

IF NO ROUTER HANDS OUT ADDRESSES

The Scout falls back to a fixed address: **192.168.100.100**. To reach it, set your laptop's Ethernet address manually to **192.168.100.10** and connect straight through the injector.

BonjourBrowser 1.15



SCAN



Dronetag SCOUT D1031854C53D1B8BD0
http://192.168.1.61:80



Dronetag SCOUT D1031854C53D1B8BD0
http://192.168.1.61:80



Dronetag SCOUT D1031854C53D1B8BD0
http://192.168.1.61:80



Dronetag SCOUT D1031854C53D1B8BD0
http://192.168.1.61:80

STEP 3

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LOG IN TO THE MANAGEMENT PAGE

- 01 On the laptop, open a browser and type `https://` followed by the Scout's IP address — for example `https://192.168.1.42`.
- 02 Your browser will show a security warning. **This is normal** — the Scout uses its own self-signed certificate. Click **Advanced** → **Proceed** to continue.
- 03 Log in with the default username and password. They are printed on the **sticker on the bottom of the Scout**.
- 04 Go to the **System** tab and change the password: enter the current one, type the new one twice, click **Update Password**.
- 05 Store the new password in the company password manager. A factory reset is the only way back if it's lost — and that wipes your settings too.

GOOD TO KNOW

This page is the Scout's control room: diagnostics, network settings, and configuration all live here. The Dronetag App shows data — but settings can only be changed here.

Forwarding

Heartbeat

Networking

GNSS

System

Networking

Static IP on ethernet ☒

Status unavailable
Address: 192.168.1.61/24
Gateway: 192.168.1.1
DNS: 192.168.1.1

In case you cannot rely on DHCP server for getting your IP address and gateway, set static IP addresses for your ethernet device end0 here.

Address

Gateway:

DNS

Add another DNS server (via its IP) to the current set of DNS servers.

UPDATE NETWORKING

STEP 4

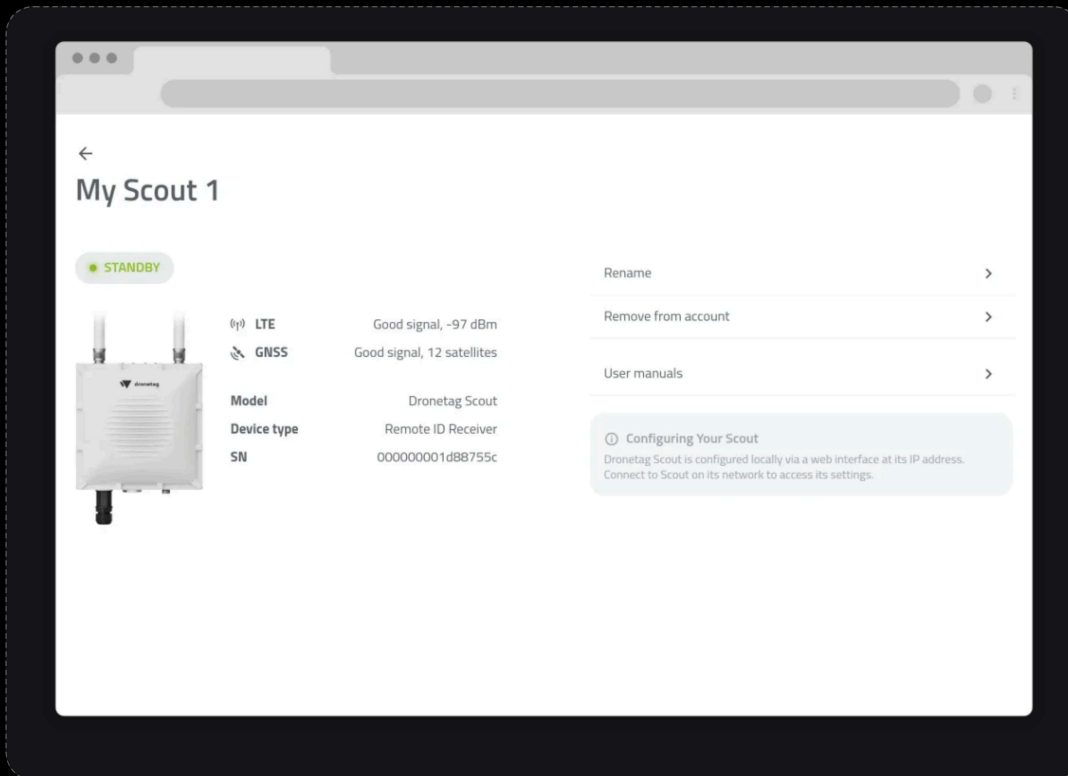
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REGISTER IT IN THE DRONETAG APP

- 01 Go to dronetag.app in a browser (or install the Dronetag App from the App Store / Google Play) and sign up for a free account — unless DXD already has one for this site.
- 02 Open **My Devices** and click **Register new** in the top-right corner.
- 03 Enter the Scout's **serial number** (on the device sticker and the box) and confirm.
- 04 In your profile, open **Cloud Data Visibility** and check the setting — it decides whether your drone data is public or private. For client sites: **private**.
- 05 Select the Scout under My Devices — you should see it online with a status overview.

PRE-REGISTERED KIT?

Some Scout kits arrive already registered to a preconfigured account — in that case Dronetag supplies the login name and password and you can skip the registration steps.



STEP 5

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PICK THE SPOT

The Scout listens with radio ears. Height and open sky are everything — it cannot hear through buildings or large obstacles.

- 01 **Go high.** The higher the Scout sits, the farther it hears. Best case: the tallest rooftop around, with a clear line of sight in all directions.
- 02 **Stay away from noise.** Keep distance from Wi-Fi routers, other antennas, mobile hotspots — anything transmitting on 2.4 or 5 GHz. They shout over the drone signals the Scout is trying to hear.
- !! **Never on the same roof as the DJI RTK Relay.** The Scout and the Relay need serious separation — put them on different rooftops or as far apart as the site allows.
- 03 **Check the pole.** The clamp fits poles 45–110 mm across. Measure before you climb.
- 04 **Plan the cable run.** The outdoor Ethernet cable must reach from the Scout down to the NEMA enclosure — or to the customer's indoor PoE cable, if they provide one.

NOT SURE ABOUT THE LOCATION?

Dronetag support will verify a site for free — send them the address or rooftop photos and they check signal conditions and interference using satellite data.



STEP 6

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MOUNT IT ON THE POLE

- 01 Place the **two clamp pieces** around the pole and tighten them with the **M6×110 bolts, washers, and nuts**. Use your two 10 mm wrenches — one holds the nut, one turns the bolt.
- 02 Secure the **cross bracket** (the last piece of the mounting kit) with the **M6×30 bolts, washers, and nuts**.
- 03 Fix the Scout so the **antennas point straight up** — perpendicular to the horizon. Tilted antennas shrink the listening range.
- 04 Push and wiggle it. Nothing should move. If it wobbles, tighten again.



MAST MOUNTING — HARDWARE REFERENCE

1 RADIO UNIT MAST MOUNTING ASSEMBLY

FEATURES

- Rugged IP-rated enclosure
- Integrated heat sink fins
- Multiple RF / power ports
- External grounding point
- Mast mount interface

ORIENTATION
UP
FRONT
RIGHT

1 TOP PORTS
RF / POWER

2 GROUND POINT

3 MOUNTING BRACKET

4 MOUNTING BOLTS (4X)

5 CABLE GLAND
POWER / DATA

MOUNTING BRACKET

SPECIFICATIONS

DIMENSIONS (H x W x D)
270 x 190 x 85 mm

WEIGHT
3.2 kg

INGRESS PROTECTION
IP67

OPERATING TEMP.
-40°C to +65°C

MOUNTING
Mast / Pole

2 MAST MOUNTING EXPLODED ASSEMBLY

COMPONENTS

- RADIO UNIT
- MOUNTING BRACKET
- PIVOT ARM
- MAST CLAMP ASSEMBLY
- HEX BOLT (M10) (2X)
- WASHER (2X)
- LOCK WASHER (2X)
- NUT (2X)

ASSEMBLY NOTES

- Use M10 hardware and tighten to 20-25 N-m
- Ensure pivot arm moves freely before final torque
- Connect ground point to mast grounding system

3 MAST CLAMP ASSEMBLY EXPLODED VIEW

COMPONENTS

- CLAMP BODY (LEFT)
- CLAMP BODY (RIGHT)
- RUBBER LINER
- HEX BOLT (M10) (2X)
- WASHER (2X)
- LOCK WASHER (2X)
- NUT (2X)

CLAMP SPECIFICATIONS

MAST DIAMETER
Ø 30 - Ø 80 mm

MAX LOAD
75 kg

MATERIAL
Die-cast Aluminum
Stainless Hardware

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STEP 7

CONNECT AND SEAL THE CABLE

Water is the number-one killer of outdoor installs. This step is all about keeping it out.

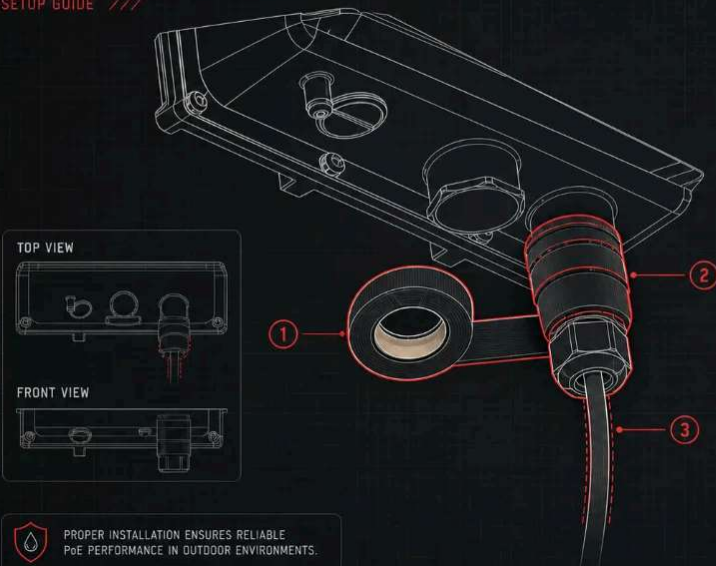
- 01 Use **outdoor-rated CAT5 or better** Ethernet cable. Indoor cable cracks in the sun within a year.
- 02 Thread the bare cable end **through the cable gland first**, then crimp the RJ-45 plug on. A crimped plug will not fit through the gland — order matters.
- 03 Click the plug into the Scout and screw the gland tight so it grips the cable jacket.
- 04 Run the cable down the pole (zip-tie every 30–50 cm) to the NEMA enclosure and into the injector's **PoE / OUT** port. Injector's LAN port to the router inside the box, same as the bench test. If the customer provided a dedicated PoE cable from indoors, connect to that instead.

TAPE THE CONNECTION — DRONETAG'S EXACT METHOD

- A Clean the surface of the connector you are about to wrap.
- B Wrap one layer of the included PVC tape with a **50% overlap**, following the direction the connector rotates.
- C Keep wrapping to about **10 mm below the end of the connector**.
- D Add **three more layers**, letting the tape uncoil naturally, still at 50% overlap.

WATERPROOF PoE CONNECTOR

SETUP GUIDE ///



1 PVC TAPE



Wrap PVC tape tightly around the connector body and cable entry to create a moisture barrier and secure the connection.

RECOMMENDED:
UV-RESISTANT PVC TAPE

2 WATERPROOF PoE CONNECTOR



Seal and protect the network connection from water, dust, and corrosion. Ensure the connector is fully tightened for a secure fit.

RATING:
IP67 WEATHERPROOF

3 CABLE



Route the Ethernet cable through the connector. Ensure proper slack and strain relief to prevent pull or tension.

COMPATIBLE:
CAT5e / CAT6 / CAT6A



IMPORTANT:

Verify all connections are sealed and secure before exposure to outdoor conditions.

STEP 8

8

POWER UP AND PROVE IT

- 01 Plug the injector into the wall. Wait 1–2 minutes for the Scout to boot.
- 02 On the laptop, open <https://<SCOUT-IP>/map> — the Scout's own quick map of every Remote ID drone it currently hears.
- 03 Fly a test drone nearby. It must have a **GPS fix and motors running** before it starts broadcasting Remote ID — some models only broadcast after takeoff.
- 04 Open the **Dronetag App** — the same drone should appear on the cloud map. That's the whole chain working: drone → Scout → cloud → app.

DURING THE TEST

Keep phones, smartwatches, hotspots, and drone controllers away from the Scout — they use the same 2.4 / 5 GHz frequencies it is listening on and can drown out the test drone.

PASS 1

SCOUT ONLINE

Shows connected in the app's My Devices.

PASS 2

DRONE ON /MAP

Test drone visible on the Scout's local map.

PASS 3

DRONE IN APP

Same drone live on the Dronetag App map.

TROUBLESHOOTING

CAN'T OPEN THE MANAGEMENT PAGE

- Is the laptop on the **same network** as the Scout? Same router, not a guest network.
- Check the router port LEDs: **green** = link, **blinking amber** = data moving, **nothing** = bad cable or dead port.
- No DHCP? Try the fallback address **192.168.100.100** with your laptop set to **192.168.100.10**.
- Some switches mis-negotiate with the Scout — manually set the port to **100 Mbit Full Duplex**.

DRONE ISN'T DETECTED

- Is the drone actually broadcasting Remote ID? It needs a **GPS fix and motors running**; some only start after takeoff.
- Move phones, hotspots, and controllers away from the Scout during the test.
- Older or sub-250 g drones may not broadcast Remote ID at all — no signal, no detection.

LOCKED OUT / BROKE THE NETWORK SETTINGS

- A **factory reset** restores default settings and the default password (on the bottom sticker).
- After reset, the Scout asks for an address via DHCP again, falling back to **192.168.100.100**.

STUCK? CALL BEFORE YOU IMPROVISE.

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HANDOFF CHECKLIST

Run through this list before you leave the site. Every item true means the install is done.

☐

All parts verified against the What You Need list

☐

Bench test passed indoors before mounting

☐

Default password changed and stored in the password manager

☐

Scout registered in the Dronetag App; data visibility set to private

☐

Mounted solid, antennas pointing straight up, no wobble

☐

Connector taped per the four-layer method; gland tight; NEMA enclosure sealed and powered (or indoor injector dry)

☐

Test drone detected on /map and in the Dronetag App

